ORIGINAL RESEARCH · SYSTEMATIC REVIEW

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EFFICACY AND SAFETY OF PUSH-DOSE NOREPINEPHRINE FOR INTRAOPERATIVE HYPOTENSION MANAGEMENT IN NON-OBSTETRIC SURGERY: A SYSTEMATIC REVIEW

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BACKGROUND & OBJECTIVES

Intraoperative hypotension (IOH, MAP <65 mmHg) occurs in ≤30% of non-cardiac surgeries and is associated with myocardial injury, AKI, and 30-day mortality. Push-dose (bolus) norepinephrine (NE) has emerged as a practical vasopressor for acute IOH. This systematic review evaluates the **efficacy and safety of push-dose NE** compared to other vasopressors in non-obstetric surgery.

P · Adults, Non-Obstetric Surgery | I · Push-Dose NE 10 mcg IV | C · Ephedrine / NE Continuous Infusion | O · MAP Target · AE Profile | Design · PRISMA 2020 · 3 RCTs

METHODS

- **Databases:** PubMed, Scopus, ProQuest
- **Keywords:** "Norepinephrine" AND ("push dose" OR "bolus") AND ("intraoperative hypotension" OR "perioperative" OR "surgery")
- **Inclusion:** Adults ≥18 yrs, non-obstetric surgery, push-dose/bolus NE for IOH, RCT
- **Exclusion:** ICU/sepsis, obstetric, case reports, narrative reviews
- **Quality:** RoB 2 (Cochrane) | **Language:** English

PRISMA FLOW

643 Records identified › **363** After dedup (−280 dupl.) › **5** Full-texts assessed › **3** RCTs included

Excluded (n=2): scoping review (Berkenbush 2024); narrative review (Duarte-Medrano 2026)

CHARACTERISTICS OF INCLUDED STUDIES

Study	n	Population	Comparator	IOH Def.	BP Monitor
Hassani 2018 (Iran)	56	Hypertensive ASA 1–2; spinal surgery GA	Ephedrine 5 mg bolus	MAP <60	Invasive arterial
Thomsen 2026 (INDUCT, Germany)	261	ASA ≥2, ≥45 yrs; elective noncardiac low–mod risk	NE continuous infusion	MAP <65	Oscillometric + CNAP
Vokuhl 2025 (Germany)	71	ASA II–IV, ≥45 yrs; elective major noncardiac high-risk	NE continuous infusion	MAP <65	Invasive arterial

All studies: NE 10 mcg/bolus · concentration 10 mcg/mL · peripheral IV route

RESULTS – EFFICACY

Study	Key Outcome	NE Bolus	Comparator	p
Hassani 2018 vs. Ephedrine	End-anaesthesia MAP (mmHg±SD)	**98.9±12.7**	91.5±10.9	<0.001
	Hypotension episodes / Total vasopressor dose	Fewer / Less	More / More	0.005 / 0.004
Thomsen 2026 vs. NE Infusion	AUC MAP <65 (mmHg·min, IQR)	5.5 (0.5–24.5)	3.6 (0.0–16.6)	0.070 NS
	Total NE dose (mcg/kg)	0.3 (0.2–0.4)	0.9 (0.6–1.1)	<0.001
Vokuhl 2025 vs. NE Infusion	ARV-MAP (mmHg/min) — *Primary*	25±7	19±6	<0.001 (d=1.00)
	AUC MAP <65 (mmHg·min, IQR)	9 (1–16)	3 (0–18)	0.178 NS

RESULTS – SAFETY

Parameter	NE Bolus	Comparator	p
AUC >MAP 100 (Thomsen 2026)	42 (18–87)	47 (9–131)	0.390 NS
AUC >MAP 100 (Vokuhl 2025)	16 (6–53)	20 (2–72)	0.986 NS
Arrhythmia	Not reported as structured outcome in any study		
Vascular complications	None documented; peripheral IV safe (all 3 studies)		

RISK OF BIAS (ROB 2)

Study	D1	D2	D3	D4	D5	Overall
Hassani 2018	⚠️	✅	✅	✅	✅	⚠️ Some
Thomsen 2026	✅	⚠️	✅	✅	✅	⚠️ Some
Vokuhl 2025	✅	⚠️	✅	✅	✅	⚠️ Some

D1 Randomisation · D2 Deviations · D3 Missing data · D4 Outcome · D5 Reporting. No study rated high risk.

⊕⊕○○ MAP reversal GRADE Low | ⊕⊕○○ HTN overshoot GRADE Low | ⊕○○○ MAP time GRADE Very Low | — Arrhythmia No evidence

KEY FINDINGS

- **vs. Ephedrine:** NE 10 mcg bolus significantly superior in maintaining MAP (p<0.001), reducing hypotension episodes (p=0.005) and total vasopressor use (p=0.004) in hypertensive patients undergoing spinal surgery.
- **vs. NE Infusion (low–mod risk):** No significant difference in AUC hypotension (p=0.070); NE bolus used 66% less total NE dose (p<0.001).
- **vs. NE Infusion (high-risk):** Continuous infusion provided better BP stability (ARV-MAP ↓25%, Cohen's d=1.00); AUC hypotension did not differ (p=0.178).
- **Safety:** No significant overshoot hypertension difference; no arrhythmia documented; peripheral IV safe across all 3 studies.

CONCLUSION

- Push-dose NE 10 mcg (10 mcg/mL, peripheral IV) is **effective and safe** for managing IOH in non-obstetric surgery.
- **Superior to ephedrine** in hypertensive patients; comparable to continuous NE infusion for low–moderate risk patients.
- For **high-risk patients with continuous arterial monitoring**, continuous infusion may offer better haemodynamic stability.
- Evidence certainty is **low (GRADE ⊕⊕○○)** due to clinical heterogeneity and limited sample (n=388). High-quality multicenter RCTs needed.

REFERENCES (MAX 7)

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